

LIU LIU, Ph.D.

Quantitative Methodology and Applied Measurement

liuul@uga.edu | liuul6218-tech.github.io | Google Scholar | Athens, GA

ACADEMIC APPOINTMENTS

Assistant Research Scientist

Oct 2024–Present

School of Social Work, University of Georgia, Athens, GA

Lead quantitative methodology and applied measurement across interdisciplinary studies involving Bayesian multilevel modeling, psychometrics, complex surveys, and program evaluation. Develop reproducible simulation and analysis workflows, contribute to study design and grant development, and mentor graduate researchers.

Faculty Affiliate

2025–Present

Owens Institute for Behavioral Research, University of Georgia

EDUCATION

Ph.D., Measurement and Statistics

Aug 2024

University of Washington

Dissertation: Informative Variance Priors for Bayesian Multilevel Models

M.A., Educational Statistics and Research Methods

Feb 2021

University of Delaware

M.S., Statistics, Measurement, Assessment, and Research Technology

Dec 2018

University of Pennsylvania

B.A., English Education

Jul 2016

Fuyang Normal University

RESEARCH AND TEACHING INTERESTS

Methodological research: Bayesian multilevel and longitudinal modeling; prior specification and calibration for small samples; psychometrics and measurement under sparse and imbalanced data.

Applied research: Quantitative methods for measurement and program evaluation in education and social research.

Teaching interests: Multilevel modeling; Bayesian data analysis; applied measurement; quantitative research methods.

PUBLICATIONS

Accepted

Liu, L. (accepted). A tutorial on selecting, translating, and reporting variance priors in Bayesian multilevel models. *Behavior Research Methods*.

Liu, L. (accepted). Longitudinal dyadic modeling with limited numbers of dyads: Aligning estimands, information sources, and multilevel structure. *Methods in Psychology*.

Liu, L. (accepted). Mode-matched inverse gamma priors for variance components in Bayesian multilevel models. *Multivariate Behavioral Research*. <https://doi.org/10.1080/00273171.2026.2740827>

Liu, L., Okech, D., & Cody, A. (accepted). Household and child correlates of child trafficking indicators in Eastern Sierra Leone. *PLOS ONE*. <https://doi.org/10.1371/journal.pone.0347898>

Peer-Reviewed Journal Articles

Liu, L., Liu, J., Yi, H., & Okech, D. (2026). Simulation-cum-ROC fit evaluation in sparse ordinal CFA under category collapsing. *Structural Equation Modeling: A Multidisciplinary Journal*, 1–12. <https://doi.org/10.1080/10705511.2026.2691170>

Liu, L., & Dai, R. (2026). Explainable AI for predicting and interpreting mathematics achievement: A cross-national analysis of PISA 2018. *Large-scale Assessments in Education*, 14, Article 46. <https://doi.org/10.1186/s40536-026-00320-y>.

Liu, L., Okech, D., Cody, A., & Konteh, F. (2026). Human-centered child trafficking prevalence: A latent class analysis of exploitation sectors. *International Journal on Child Maltreatment: Research, Policy and Practice*, 1–30. <https://doi.org/10.1007/s42448-026-00266-2>

Okech, D., Liu, L., Cody, A., Hwang, E., & Konteh, F. (2026). Sex trafficking and commercial sex: Outcomes in suicide ideation and economic strain among young women. *Frontiers in Psychiatry*, 17, Article 1843240. <https://doi.org/10.3389/fpsy.2026.1843240>

- Yi, H., Liu, X., Okech, D., Clay-Warner, J., **Liu, L.**, & Deng, X. (2026). New perspectives on respondent-driven sampling with application to sex-trafficked population in Senegal. *Journal of Statistical Theory and Practice*, 20(3), Article 84. <https://doi.org/10.1007/s42519-026-00612-0>
- Liu, L.**, Joseph, G. E., Taylor, J. M., Hassairi, N., & Soderberg, J. S. (2025). Early childhood educators pay equity: A dream deferred. *Early Childhood Education Journal*, 53(2), 463-476. <https://doi.org/10.1007/s10643-023-01600-w>
- Berridge, C., Turner, N. R., **Liu, L.**, Fredriksen-Goldsen, K. I., Lyons, K. S., Demir, G., ... & Lober, W. B. (2023). Preliminary efficacy of Let's Talk Tech: Technology use planning for dementia care dyads. *Innovation in Aging*, 7(3), igad018. <https://doi.org/10.1093/geroni/igad018>
- Collier, Z. K., Zhang, H., & **Liu, L.** (2022). Explained: Artificial intelligence for propensity score estimation in multilevel educational settings. *Practical Assessment, Research, and Evaluation*, 27(1), 3. <https://doi.org/10.7275/0dpq-eq84>
- Berridge, C., Turner, N. R., **Liu, L.**, Karras, S. W., Chen, A., Fredriksen-Goldsen, K., & Demir, G. (2022). Advance planning for technology use in dementia care: Development, design, and feasibility of a novel self-administered decision-making tool. *JMIR Aging*, 5(3), e39335. <https://doi.org/10.2196/39335>
- Book Chapter**
- Lü, C., Pace, A. E., & **Liu, L.** (2023). Student-level variables and academic achievement in a Mandarin dual language immersion program. In D. Zhang & R. T. Miller (Eds.), *Crossing boundaries in researching, understanding, and improving language education* (pp. 213–229). Springer. https://doi.org/10.1007/978-3-031-24078-2_10

MANUSCRIPTS UNDER REVIEW AND IN REVISION

Methodological Research

- Liu, L.** (2026). Calibrating informative variance priors in Bayesian multilevel item-response models: A practical evaluation using PISA 2018. *Applied Measurement in Education* (under review).
- Liu, L.** (2026). Benchmark-based calibration of scalar variance priors across Bayesian multilevel software. *Methodology* (under review).
- Liu, L.** (2026). Role parameterization in longitudinal models for distinguishable dyads: A practical framework for fixed effects, random effects, and post-estimation. *Behavior Research Methods* (under review).
- Liu, L.** (2026). How dyad sample size and repeated measurements shape the performance of multilevel actor-partner growth models for longitudinal distinguishable dyads. *Multivariate Behavioral Research* (under review).
- Liu, L.**, & Collier, Z. (2026). A DoWhy-assisted causal workflow for observational educational evaluation: An applied charter school enrollment case. *Studies in Educational Evaluation* (under revision).
- Liu, L.**, Liu, J., & Okech, D. (2026). Calibrating fit-index cutoffs for sparse ordinal CFA: Scenario-specific thresholds and composite rules. *Behavior Research Methods* (under revision).
- Liu, J., **Liu, L.**, Jiang, W., Wang, X., & Kellam, H. (2026). When collapsing sparse categories alters model decisions: Evidence from confirmatory factor analysis across estimators. *Behavior Research Methods* (under review).

Applied Measurement and Evaluation

- Okech, D., **Liu, L.**, Cody, A., Konteh, F., & Hwang, E. M. (2026). Child abuses through force, fraud, and coercion: A latent class analysis of child trafficking in Sierra Leone. *Child Abuse Review* (under review).
- Goulart, P., Okech, D., **Liu, L.**, Cody, A., & Chunga, J. (2026). Validation of human trafficking as a latent construct using the Prevalence Reduction Innovation Forum instrument: Evidence from Zambia and Malawi. *Measurement Instruments for the Social Sciences* (under review).
- Goulart, P., Okech, D., **Liu, L.**, & Cody, A. (2026). Measurement validity of the Multidimensional Scale of Perceived Social Support in link-tracing sampling data from mobile youth and young adults in Zambia. *International Journal of Population Data Science* (under review).
- Goulart, P., Okech, D., **Liu, L.**, & Cody, A. (2026). Perceptions of social support among mobile young workers in Malawi: Evidence from link-tracing sampling data. *Journal of Ethnic and Migration Studies* (under review).
- Mbamba, C. R., Okech, D., Cody, A. M., & **Liu, L.** (2026). A three-dimensional typology validity framework and latent class analysis of child exploitation and trafficking. *Child Abuse & Neglect* (under review).
- Yang, H., Liu, J., & **Liu, L.** (2026). Impacts of university students' ethical ideology on their ethical sensitivity: A Bayesian moderation analysis. *Journal of Psychoeducational Assessment* (under revision).

STAKEHOLDER-FACING RESEARCH REPORTS

Okech, D., Kinnish, K., **Liu, L.**, Dockweiler, H., Cody, A., Hwang, E. M., & Caplan, S. (2026). *Aviation-connected human trafficking landscape and stakeholder ecosystem in Atlanta, Georgia*. University of Georgia Center on Human Trafficking Research and Outreach.

Okech, D., Kinnish, K., Cody, A., Dockweiler, H., & **Liu, L.** (2026, July). *Preliminary stakeholder perspectives on human trafficking and aviation: Implications for awareness, identification, response, and collaboration* [Technical report]. University of Georgia Center on Human Trafficking Research and Outreach.

Okech, D., Cody, A. M., Clay-Warner, J., Balch, A., Yi, H., Callands, T., Bolton, C., Konteh, F., Jalloh, A., Dockweiler, H., Goulart, P., Schroeder, E., **Liu, L.**, Eng, C., Waswa, A., & Fofanah, U. (2025). *Assessing changes in child trafficking and the worst forms of child labor in Sierra Leone from 2019–2024*. University of Georgia Center on Human Trafficking Research and Outreach. <https://doi.org/10.71927/UGA.27593>

GRANTS AND SPONSORED PROJECTS

Funded

Principal Investigator. *Bayesian Multilevel Modeling for Social Work Research*. University of Georgia School of Social Work Seed Grant, 2025. \$5,600.

Co-Investigator. *Delta Air Lines-University of Georgia Center on Human Trafficking Research Collaboration Project*. University of Georgia, 2025. \$80,000.

Statistical Consultant. *REBOOT Award*. New York University, 2025. Longitudinal cohort study of health, social, and economic trajectories among adolescents and young adults affected by commercial sexual exploitation.

In Preparation

Subaward Principal Investigator. *Computational Experiences in Introductory Mathematics: A Longitudinal Study of Mathematical Agency and STEM Progression*. National Science Foundation ECR:Core Level II, 2026. Proposal in preparation.

Submitted and Collaborative

Co-Investigator. *A Systems- and Human-Centered Prevention Intervention to Reduce Child Trafficking in Malawi*. Program to End Modern Slavery, Office to Monitor and Combat Trafficking in Persons, U.S. Department of State, 2026. Under review.

Co-Investigator. *Estimating the Impacts of Supplemental Nutrition Assistance Program on Child Health: A Causal and Computational Approach*. National Institutes of Health R03, 2025. \$151,000 requested. Submitted.

TEACHING, MENTORING, AND PROFESSIONAL EDUCATION

Invited Guest Lecturer, SOWK 8307 Doctoral Research Practicum II <i>School of Social Work, University of Georgia</i> Delivered a doctoral seminar on the quantitative research process, including research questions, study design, measurement, data analysis, interpretation, and responsible reporting.	Apr 2026
Teaching Assistant, EDPSY 576 Multilevel Modeling <i>University of Washington</i> Led weekly R-based laboratories on model specification, diagnostics, interpretation, and reproducible reporting; provided individual methodological consultation to more than 30 graduate students.	Sep–Dec 2023
Research Methods Mentor and Statistical Consultant <i>Independent and collaborative research teams</i> Mentor graduate students and research trainees on study design, measurement, analysis planning, model selection, diagnostics, interpretation, and reproducible documentation.	Sep 2020–Present

SELECTED RESEARCH AND APPLIED MEASUREMENT EXPERIENCE

Graduate Research Assistant, College of Education, University of Washington Developed Bayesian multilevel methods and reproducible simulation workflows, including variance-prior research in multilevel and item-response settings and applications to large-scale assessment data.	Jan 2022–Sep 2024
Graduate Research Assistant, Cultivate Learning, University of Washington Conducted survey design, sampling and weighting, workforce analytics, and policy-facing quantitative reporting.	Sep 2022–Mar 2024

Summer Data Science Intern, National Commission on Certification of Physician Assistants 2022 and 2023
Developed natural-language-processing and machine-learning workflows for operational certification item review, including enemy-item detection and reproducible text-analysis pipelines.

Summer Data Science Intern, Curriculum Associates Jun–Aug 2021
Conducted large-scale differential item functioning analyses on more than 10 million i-Ready computerized adaptive assessment records to evaluate item fairness across demographic groups.

Graduate Research Assistant, School of Social Work, University of Washington Sep 2021–Jun 2022
Supported survey sampling and weighting, data management, and multilevel dyadic analyses in child and family research.

SELECTED CONFERENCE PRESENTATIONS AND WORKSHOPS

- Liu, L., Okech, D., Cody, A., & Goulart, P. (2027, January). *Are human trafficking severity scores comparable across contexts? Evidence from Malawi and Zambia* [Accepted paper presentation]. *Society for Social Work and Research Annual Conference*, San Francisco, CA.
- Yang, H., Liu, J., & Liu, L. (2026, October). *A Bayesian approach to understanding academic integrity: How ethical ideology shapes student sensitivity* [Accepted paper presentation]. Mid-Western Educational Research Association Annual Meeting, Milwaukee, WI.
- Yi, H., Hasan, A., Liu, L., Okech, D., & Olalere, A. (2026, October). *An explainable artificial intelligence framework for child trafficking risk assessment* [Accepted paper presentation]. Fall Technical Conference, Charlotte, NC.
- Liu, L., Liu, J., & Okech, D. (2026, June). *Scenario-specific fit-index cutoffs for sparse ordinal CFA: Monte Carlo calibration across estimators and category collapsing* [Paper presentation]. Modern Modeling Methods Conference, New York, NY.
- Okech, D., Liu, L., Cody, A., Bolton, C., Yi, H., Waswa, A., & Goulart, P. (2026, June). *Measuring the unseen: Mixed methods and Bayesian multilevel modelling in survivor-centered child trafficking research* [Workshop]. The Joint Conference on Social Work, Education and Social Development, Nairobi, Kenya.
- Liu, J., Liu, L., & Jiang, W. (2026, April). *Assessing the performance of Bayesian and traditional estimators in confirmatory factor analysis with sparse Likert-scale data* [Paper presentation]. American Educational Research Association Annual Meeting, Los Angeles, CA.
- Liu, L., Cody, A., Okech, D., Yi, H., Waswa, A., & Goulart, P. (2026, April). *Beyond rescue: Child trafficking, educational access and reintegration in Eastern Sierra Leone* [Paper presentation]. American Educational Research Association Annual Meeting, Los Angeles, CA.
- Liu, L., Okech, D., & Yi, H. (2025, August). *Informative variance priors for Bayesian multilevel modeling with large and small samples* [Paper presentation]. Joint Statistical Meetings, Nashville, TN.
- Liu, L., & Sanders, E. A. (2024, April). *Informative inverse-gamma variance priors for Bayesian multilevel item response model precision* [Paper presentation]. National Council on Measurement in Education Annual Meeting, Philadelphia, PA.
- Liu, L., & Walker, M. (2024, April). *Fine-tuned word embeddings for enemy item detection* [Paper presentation]. National Council on Measurement in Education Annual Meeting, Philadelphia, PA.
- Liu, L., Walker, M., & Weir, J. (2023, November). *Unveiling enemy item pairs through machine learning approaches* [Paper presentation]. Pacific Northwest Conference for Research on Psychometrics & Applied Statistics, Seattle, WA.
- Liu, L., & Sanders, E. A. (2023, July). *Bayesian variance component priors for small-sample multilevel models* [Poster]. International Meeting of the Psychometric Society, College Park, MD.
- Liu, L., & Sanders, E. A. (2023, April). *On informative Bayesian multilevel model priors for variance components* [Poster]. American Educational Research Association Annual Meeting, Chicago, IL.
- Liu, L., Walker, M., & Weir, J. (2023, April). *An exploration of natural language processing for enemy item detection* [Paper presentation]. National Council on Measurement in Education Annual Meeting, Chicago, IL.
- Liu, L., & Boyd, A. (2022, April). *DIF analyses for embedded field test items in a computerized adaptive test* [Paper presentation]. National Council on Measurement in Education Annual Meeting.
- Liu, L., & Collier, Z. (2021, April). *Causal inference made easy: A review of the DoWhy package in Python* [Paper presentation]. American Educational Research Association Annual Meeting.

PROFESSIONAL SERVICE AND LEADERSHIP

Secretary/Treasurer, AERA Special Interest Group: Multiple Linear Regression and General Linear Models, 2025–2028.

Campus Representative, AERA Graduate Student Council, 2020–2023.

Consulting Editor, *Teaching Educational Research Methods*, 2025–Present.

Editorial Review Board, *Journal of Applied Educational Research and Evaluation (JAERE)*, 2026–Present.

Ad hoc reviewer, 40+ journal manuscripts, including *Methodology*, *Structural Equation Modeling: A Multidisciplinary Journal*, *Journal of Psychoeducational Assessment*, *Early Childhood Education Journal*, and *Education Sciences*.
Conference reviewer, 80+ submissions, including AERA, NCME, and the International Test Commission Conference.
Session Chair, AERA 2026, Multiple Linear Regression/General Linear Models SIG and Educational Statisticians SIG.
Discussant, AERA Division D, 2021; National Council on Measurement in Education, 2021.

FELLOWSHIPS, AWARDS, SPECIALIZED TRAINING, AND MEMBERSHIPS

- Translational Education and Mentoring in Science Fellow, Georgia Clinical & Translational Science Alliance, Emory University 2025–2026
- College of Education Graduate Student Scholarship, University of Washington 2021–2023
- Tolo Scholarship, Mortar Board Alumni/Tolo Foundation, University of Washington 2022
- Chester Fritz International Research Fellowship, University of Washington 2021
- National Assessment of Educational Progress Data Training Workshop Fellow, National Center for Education Statistics and American Institutes for Research 2020
- Foundry10 Research Scholarship, Foundry10 Education Research 2020
- Global Internship Program Award, University of Pennsylvania 2018
- Research Grant Award, Center for the Study of Contemporary China, University of Pennsylvania 2018
- **Professional memberships:** American Educational Research Association; American Statistical Association; National Council on Measurement in Education; Psychometric Society.

METHODS AND SOFTWARE

Methods: Bayesian multilevel and longitudinal modeling; CFA/SEM and item-response modeling; measurement invariance and DIF; complex survey analysis; simulation; causal inference; machine learning and NLP for measurement applications.
Software: R, Python, Stan, Mplus, SAS, SPSS, Winsteps, Git, LaTeX.